

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W003
Well Name	SYN-Assam-Duliajan-003
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.329164 N
Longitude	95.330380 E
Well Status	Drilled
Well Role	Historical (offset reference well)
Spud Date	10-Dec-2018
Drilling Completed	22-Dec-2018
Report / Completion Date	27-Dec-2018
Total Depth (Driller)	3,850 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-04
Drilling Days	13 days
Recorded Events	8
Recorded NPT	108.2 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W003
Well Name: SYN-Assam-Duliajan-003
Field: SYN-Assam-Duliajan
Latitude: 27.329164
Longitude: 95.330380
Total Depth: 3,850 m
Primary Formation: Sandstone
Well Status: Drilled
Spud Date: 2018-12-10
Event Count: 8
Total NPT: 108.2 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

DOC No. WCR-W003-R0 · ISSUE 27-Dec-2018 · PROTOTYPE TRAINING RECORD

3. WELL SUMMARY

3.1 Objective

Well W003 (SYN-Assam-Duliajan-003) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,850 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,850 m, completed in 13 rotary drilling days between 10-Dec-2018 and 22-Dec-2018. Average overall penetration rate for the well was 12.3 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–867 m), Shale (867–2,190 m), Sandstone (2,190–3,427 m), Limestone (3,427–3,850 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

A total of 8 reportable events were recorded, comprising 4 × Casing Issue, 1 × Stuck Pipe, 2 × Mud Loss, 1 × Cementing Issue. Aggregate non-productive time was 108.2 hours. Event detail is given in Section 6.

3.5 Casing Operations

4 casing strings were run: 13-3/8" at 865 m, 9-5/8" at 2,190 m, 7" at 3,610 m, 5" liner at 3,850 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Drilled**. The well was drilled and logged to total depth and suspended pending completion decisions.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W001	SYN-Assam-Duliajan-001	4.68	3,470	Sandstone	4
W008	SYN-Assam-Duliajan-008	4.92	3,690	Sandstone	8
W006	SYN-Assam-Duliajan-006	5.24	3,960	Sandstone	0
W005	SYN-Assam-Duliajan-005	7.45	3,560	Sandstone	5
W002	SYN-Assam-Duliajan-002	7.85	3,450	Sandstone	7
W004	SYN-Assam-Duliajan-004	9.47	3,590	Sandstone	2

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	867	867	Casing Issue recorded (1 event)
2	Shale	867	2,190	1,323	Casing Issue, Stuck Pipe recorded (4 events)
3	Sandstone	2,190	3,427	1,237	Mud Loss, Cementing Issue recorded (3 events)
4	Limestone	3,427	3,850	423	Routine drilling, no reportable event

Formation: Claystone Top: 0 m Bottom: 867 m
Formation: Shale Top: 867 m Bottom: 2190 m
Formation: Sandstone Top: 2190 m Bottom: 3427 m
Formation: Limestone Top: 3427 m Bottom: 3850 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–867	9.63	9.1–10.3	31.7	140	8.2
Shale	867–2,190	11.27	10.3–12.2	15.9	123	18.7
Sandstone	2,190–3,427	12.88	11.9–13.8	14.7	108	27.0
Limestone	3,427–3,850	14.43	14.2–14.7	7.0	86	37.8

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 14.4 ppg at total depth. Mud weight was trimmed to the minimum safe value across the recorded loss intervals and lost-circulation material was maintained in the active system.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 118 rpm range on average with weight on bit averaging 21.2 klbf. Penetration rate declined with depth from 31.7 m/hr in the upper hole to 7.0 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

All reportable events for this well, in depth order. Each event is recorded against the formation interval containing the event depth.

Event ID	Depth (m)	Formation	Event Type	Severity	Duration (hr)
E0030	612	Claystone	Casing Issue	Low	6.6
E0027	958	Shale	Casing Issue	Low	5.6
E0031	1,049	Shale	Casing Issue	Medium	14.2
E0033	1,169	Shale	Casing Issue	Critical	45.6
E0032	1,447	Shale	Stuck Pipe	Medium	14.1
E0028	2,210	Sandstone	Mud Loss	Medium	6.2
E0003	2,840	Sandstone	Mud Loss	High	12.3
E0029	2,987	Sandstone	Cementing Issue	Low	3.6

Well ID: W003		Event Type: Casing Issue		Depth: 612 m		Formation: Claystone		Severity: Low		Duration: 6.6 hours
Well ID: W003		Event Type: Casing Issue		Depth: 958 m		Formation: Shale		Severity: Low		Duration: 5.6 hours
Well ID: W003		Event Type: Casing Issue		Depth: 1049 m		Formation: Shale		Severity: Medium		Duration: 14.2 hours
Well ID: W003		Event Type: Casing Issue		Depth: 1169 m		Formation: Shale		Severity: Critical		Duration: 45.6 hours
Well ID: W003		Event Type: Stuck Pipe		Depth: 1447 m		Formation: Shale		Severity: Medium		Duration: 14.1 hours
Well ID: W003		Event Type: Mud Loss		Depth: 2210 m		Formation: Sandstone		Severity: Medium		Duration: 6.2 hours
Well ID: W003		Event Type: Mud Loss		Depth: 2840 m		Formation: Sandstone		Severity: High		Duration: 12.3 hours
Well ID: W003		Event Type: Cementing Issue		Depth: 2987 m		Formation: Sandstone		Severity: Low		Duration: 3.6 hours

7.1 EVENT DETAIL (1–3 OF 8)

EVENT E0030 — CASING ISSUE AT 612 m (CLAYSTONE)

Well ID: W003
Event ID: E0030
Event Type: Casing Issue
Depth: 612 m
Formation: Claystone
Severity: Low
Duration: 6.6 hours
Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Restricted casing passage through plastic Claystone at 612 m; hole conditioned and casing rerun. Parameters recorded around the event depth: mud weight 9.91 ppg, ROP 30.3 m/hr, 139 rpm, WOB 9.7 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

EVENT E0027 — CASING ISSUE AT 958 m (SHALE)

Well ID: W003
Event ID: E0027
Event Type: Casing Issue
Depth: 958 m
Formation: Shale
Severity: Low
Duration: 5.6 hours
Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment.

Description. Restricted casing passage through plastic Shale at 958 m; hole conditioned and casing rerun. Parameters recorded around the event depth: mud weight 10.54 ppg, ROP 17.4 m/hr, 135 rpm, WOB 15.5 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

EVENT E0031 — CASING ISSUE AT 1,049 m (SHALE)

Well ID: W003
Event ID: E0031
Event Type: Casing Issue
Depth: 1049 m
Formation: Shale
Severity: Medium
Duration: 14.2 hours
Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Casing integrity concern noted opposite Shale at 1049 m; pressure test anomaly investigated before proceeding. Parameters recorded around the event depth: mud weight 10.57 ppg, ROP 17.3 m/hr, 133 rpm, WOB 15.6 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

7.2 EVENT DETAIL (4–6 OF 8)

EVENT E0033 — CASING ISSUE AT 1,169 m (SHALE)

Well ID: W003
 Event ID: E0033
 Event Type: Casing Issue
 Depth: 1169 m
 Formation: Shale
 Severity: Critical
 Duration: 45.6 hours
 Operational Impact: Major non-productive time; section objectives affected and specialist intervention required.

Description. Restricted casing passage through plastic Shale at 1169 m; hole conditioned and casing rerun. Parameters recorded around the event depth: mud weight 10.71 ppg, ROP 16.1 m/hr, 127 rpm, WOB 15.5 klbf.

Response. Casing run suspended, string pulled back and the hole conditioned/reamed over the restriction before the casing was rerun; pressure test repeated after landing.

Outcome. Casing landed at the planned shoe depth and tested satisfactorily. Running speed limited through the affected interval on subsequent strings.

Historical observation. Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.

EVENT E0032 — STUCK PIPE AT 1,447 m (SHALE)

Well ID: W003
 Event ID: E0032
 Event Type: Stuck Pipe
 Depth: 1447 m
 Formation: Shale
 Severity: Medium
 Duration: 14.1 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Drill string became stuck at 1447 m within the Shale section; overpull observed and jarring / working of pipe required. Parameters recorded around the event depth: mud weight 11.17 ppg, ROP 15.0 m/hr, 120 rpm, WOB 18.7 klbf.

Response. String worked and jarred within the allowable overpull; lubricant pill spotted and hole cleaning improved with high-viscosity sweeps and increased rotation before back-reaming.

Outcome. String freed and circulation restored. Hole conditioned over the affected interval and reaming carried out before drilling ahead.

Historical observation. Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.

EVENT E0028 — MUD LOSS AT 2,210 m (SANDSTONE)

Well ID: W003
 Event ID: E0028
 Event Type: Mud Loss
 Depth: 2210 m
 Formation: Sandstone
 Severity: Medium
 Duration: 6.2 hours
 Operational Impact: Partial non-productive time; drilling suspended while the well was treated and conditioned.

Description. Loss of circulation into porous Sandstone at 2210 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 11.95 ppg, ROP 19.7 m/hr, 115 rpm, WOB 21.5 klbf.

Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet.

Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval.

Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.

7.3 EVENT DETAIL (7–8 OF 8)

EVENT E0003 — MUD LOSS AT 2,840 m (SANDSTONE)
Well ID: W003 Event ID: E0003 Event Type: Mud Loss Depth: 2840 m Formation: Sandstone Severity: High Duration: 12.3 hours Operational Impact: Significant non-productive time; remedial operations and additional materials required. Description. Loss of circulation into porous Sandstone at 2840 m; drilling paused and a lost-circulation-material pill spotted. Parameters recorded around the event depth: mud weight 12.79 ppg, ROP 14.8 m/hr, 111 rpm, WOB 26.0 klb. Response. Pumps slowed and a lost-circulation-material pill (fibrous/granular blend) spotted across the loss zone; mud weight trimmed to the minimum safe value and ECD reduced; pit levels monitored on a dedicated loss sheet. Outcome. Returns re-established and losses reduced to seepage. Drilling resumed with reduced flow rate and controlled ROP through the remainder of the interval. Historical observation. Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
EVENT E0029 — CEMENTING ISSUE AT 2,987 m (SANDSTONE)
Well ID: W003 Event ID: E0029 Event Type: Cementing Issue Depth: 2987 m Formation: Sandstone Severity: Low Duration: 3.6 hours Operational Impact: Minor non-productive time; operations resumed after routine parameter adjustment. Description. Cement slurry losses while cementing across Sandstone at 2987 m; top of cement lower than designed. Parameters recorded around the event depth: mud weight 13.14 ppg, ROP 15.1 m/hr, 103 rpm, WOB 27.7 klb. Response. Job evaluated against the pump/returns record; remedial squeeze / top-up cementing carried out and a cement bond log (CBL/VDL) run to confirm coverage. Outcome. Acceptable bond and annular isolation confirmed after the remedial job. Slurry design and displacement rate revised for the next stage. Historical observation. Annular coverage is sensitive to losses and centralisation across the shoe interval.

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	865	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	2,190	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 750 m
7"	N-80, 29 lb/ft	3,610	8-1/2"	Tail 15.8 ppg, TOC 2,040 m
5" liner	N-80, 18 lb/ft	3,850	6-1/8"	Tail 15.8 ppg, liner lap tested

8.1 Cementing Observations

The following casing/cementing events were recorded and are cross-referenced to Section 7:

- **E0030** — Casing Issue at 612 m in Claystone (Low, 6.6 hr). Restricted casing passage through plastic Claystone at 612 m; hole conditioned and casing rerun.
- **E0027** — Casing Issue at 958 m in Shale (Low, 5.6 hr). Restricted casing passage through plastic Shale at 958 m; hole conditioned and casing rerun.
- **E0031** — Casing Issue at 1,049 m in Shale (Medium, 14.2 hr). Casing integrity concern noted opposite Shale at 1049 m; pressure test anomaly investigated before proceeding.
- **E0033** — Casing Issue at 1,169 m in Shale (Critical, 45.6 hr). Restricted casing passage through plastic Shale at 1169 m; hole conditioned and casing rerun.
- **E0029** — Cementing Issue at 2,987 m in Sandstone (Low, 3.6 hr). Cement slurry losses while cementing across Sandstone at 2987 m; top of cement lower than designed.

Remedial work was carried out as described in Section 7 and annular isolation was confirmed before drilling ahead.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,850 m
Primary Formation	Sandstone
Major Events	2 of 8 events rated High or Critical
Total NPT	108.2 hours
Completion Status	Drilled
Offset Risk Classification	HIGH — repeated significant events

9.1 Drilling Challenges

- **Casing Issue** at 612 m, 958 m, 1,049 m, 1,169 m (Claystone, Shale) — 72.0 hr cumulative.
- **Stuck Pipe** at 1,447 m (Shale) — 14.1 hr cumulative.
- **Mud Loss** at 2,210 m, 2,840 m (Sandstone) — 18.5 hr cumulative.
- **Cementing Issue** at 2,987 m (Sandstone) — 3.6 hr cumulative.

9.2 Key Historical Observations

- Tight/plastic intervals reduce casing running clearance; conditioning trips before running casing are effective.
- Hole-cleaning efficiency and time spent stationary are the dominant contributors to sticking tendency in this section.
- Loss-zone behaviour is depth- and formation-controlled in this area; the porous sand intervals accept fluid once ECD exceeds the fracture/pore window.
- Annular coverage is sensitive to losses and centralisation across the shoe interval.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- Allow a dedicated conditioning trip before running casing across 612–1,169 m and limit running speed through the tight interval.
- Improve hole cleaning and minimise stationary time between 1,447 and 1,447 m; plan a wiper trip before extended surveys in this interval.
- Approach the Sandstone interval (2,210–2,840 m) with LCM pre-mixed and the minimum safe mud weight; monitor returns continuously through this window.
- Review slurry design, centralisation and displacement rate for the stage covering 2,987 m; run a bond log before proceeding.

10. DOCUMENT CONTROL

Document Number	WCR-W003-R0
Revision	R0
Issue Date	27-Dec-2018
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv
Prototype Training Record · Synthetic demonstration data	